MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M78E1
PARTS AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of electric detonator designated as Detonator, Electric, M78E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation Method of
MIL-B-117	-	Bag, Interior Packaging
MIL-A-148	-	Aluminum Foil
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables For Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9248640 - Detonator, Electric:
R.F. M78E1 Assembly
9248649 - R.F. Filter & Plug Assembly

PUBLICATIONS

U.S. ARMY MUNITIONS COMMAND

ET 9248640 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply. Code of Federal Regulations, Title 49 - Transportation, Parts 0-190.

(The Code of Federal Regulations is available from the superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders shall specify 49 CFR 0-190 (latest revision).)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Detonator assembly.-The detonator assembly shall comply with all requirements specified on Drawing (dwg.) 9248640 and with all requirements specified in applicable specifications.

3.3 Moisture content

3.3.1 Lead azide.-At the time of loading, the moisture content of the lead azide used in loading the detonators shall not exceed 0.50 percent when tested as specified in 4.4.1.1.

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3.3.2 Pentaerythrite tetranitrate (PETN).-At the time of loading, the moisture content of the PETN used in loading the detonators shall not exceed 0.50 percent when tested as specified in 4.4.1.2.

3.4 Functioning.-The detonator assembly shall comply with the following:

3.4.1 Time.-The detonator assembly shall fire on the first attempt within one hundred and twenty (120) microseconds when tested as specified in 4.4.7.1.

3.4.2 Output.-The detonator assembly shall fire on the first attempt and shall make a 3/16 inch minimum (min.) diameter perforation in a .1345 inch thick lead disc when tested as specified in 4.4.7.2.

3.4.3 Detonator that fails to fire on the first attempt (Duds).-The detonator assembly shall not fail to fire on the first attempt when tested as specified in 4.4.7.3.

3.5 Waterproofness.-The detonator assembly shall comply with the requirements of 3.4.2 and 3.4.3 when tested as specified in 4.4.7.2.

3.6 Environmental

3.6.1 Temperature and humidity.-The detonators shall comply with 3.4.1 when tested as specified in 4.4.10.1.

3.6.2 Lead circuit resistance, thermal shock, and vibration.-The detonators shall comply with the lead circuit resistance requirement of dwg. 9248640 and with 3.4.1 when tested as specified in 4.4.10.2.

3.6.3 Cold temperature.-The detonator assembly shall comply with 3.4.3 when tested as specified in 4.4.10.3.

3.6.4 Hot tmperature.-The detonator assembly shall comply with 3.4.3 when tested as specified in 4.4.10.4.

3.6.5 Jolt test.-The detonator assembly shall not function during the test, and shall be safe to handle after the test specified in 4.4.10.5.

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3.7 Workmanship.-All parts shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.8 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production, the contractor shall submit a frist article sample (see 6.1c) to a Government approved facility as designated by the contracting officer for evaluation as specified herein. The first article sample shall consist of inert parts and subassemblies (see 4.2.4) and loaded detonators (see 4.2.5), which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inert parts.-A sample of twenty-five (25) each of every inert component parts and subassemblies shall be submitted in accordance with the provisions of 4.2.1 for inspection as specified in 4.2.3.

4.2.2.1 Inspections to be performed.-The component parts and subassemblies including the R.F. Filter and Plug Assembly will be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements in the applicable drawings.

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4.2.2.2 Rejection.-If any component part or subassembly fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part to comply with any of the stated requirements.

4.2.3 Loaded detonators.-A sample of three hundred thirty four (334) loaded detonators shall be submitted in accordance with the provisions of 4.2.1 for testing as specified in 4.2.3.1.

4.2.3.1 Tests.-The detonators shall be subjected to the following tests:

4.2.3.1.1 Time (see 3.4.1), output (see 3.4.2), duds (see 3.4.3), and waterproofness (see 3.5).-One hundred and eight-five (185) detonator assemblies shall be selected and tested as specified in 4.3.3.9, Table II.

4.2.3.1.3 Temperature and humidity (see 3.6.1).-Thirty-seven (37) detonator assemblies shall be selected and tested as specified in 4.4.10.1.

4.2.3.1.3 Lead circuit resistance, thermal shock and vibration (see 3.6.2).-Thirty-seven (37) detonator assemblies shall be selected and tested as specified in 4.4.10.2.

4.2.3.1.4 Cold temperature conditioning (see 3.6.3).-Twenty-five (25) detonator assemblies shall be tested as specified in 4.4.10.3.

4.2.3.1.5 Hot temperature conditioning (see 3.6.4).-Twenty-five (25) detonator assemblies shall be tested as specified in 4.4.10.4.

4.2.3.1.6 Jolt (see 3.6.5).-Twenty-five (25) detonator assemblies shall be selected from the first article sample for this test. The test shall be performed as specified in 4.4.10.5. These samples shall not be tested for functioning.

4.2.3.2 Rejection.-The loaded detonator first article sample shall be rejected if any of the following test results occur:

4.2.3.2.1 Time, output, duds, and waterproofness.-The rejection criteria of 4.3.3.9.2 shall apply.

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4.2.3.2.2 Temperature and humidity.-The rejection criteria of 4.3.3.9.2 sub-paragraph "a" shall apply.

4.2.3.2.3 Lead circuit resistance, thermal shock and vibration.-The total number of detonator assemblies that fail to comply with the lead circuit resistance requirement equals or exceed the requirement (see 4.3.3.10.2a).

4.2.3.2.4 Cold temperature conditioning.-One or more detonator assemblies fails to fire.

4.2.3.2.5 Hot temperature conditioning.-One or more detonator assemblies fails to fire.

4.2.3.2.6 Jolt.-Any detonator assemblies that fails to meet the requirements specified in 3.6.5 when tested as specified in 4.4.10.5. The Government reserves the right to terminate its inspection upon any failure of an item to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with the applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonator assemblies shall contain:

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a. Parts of one lot interfix number from one supplier.
b. PETN from not more than one lot.
c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects uisng an AQL of 0.40 percent for each major defect and the AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 9243862 covering a detail of dwg. 9248640).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Thickness through bottom	Gage	01001
102.	Total length	Gage	01002
103.	Dent or distortion	Visual	01003
104.	Foreign matter on interior surface	Visual	01004
Minor:	AQL 2.50 percent		
201.	Profile of lip	Gage	01005
202.	Inside diameter, min	Gage	01006
203.	Outside diameter (maximum) max	Gage	01007
204.	Radius missing	Visual	01008
205.	Evidence of poor workmanship (see 3.7)	Visual	01009

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4.3.2.2 Chip, insulator (see dwg. 9248656 covering a detail of dwg. 9248649).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Silver missing or incomplete	Visual	02001
102.	Edges not free of silver	Visual	02002
Minor:	AQL 1.50 percent		
201.	Length	Gage	02003
202.	Width	Gage	02004
203.	Evidence of poor workmanship (see 3.7)	Visual	02005

4.3.2.3 Bead, Ferrite (see dwg. 9248658 covering a detail of dwg. 9248649).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Length	Gage	03001
102.	Outside diameter	Gage	03002
103.	Inside diameter	Gage	03003
104.	Metalic coating missing or incomplete	Visual	03004
105.	Metalic coating where not required	Visual	03005
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	03006

4.3.2.4 Ground strap (see dwg. 9248657 covering a detail of dwg. 9248649).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Length	Gage	04001
102.	Inside width	Gage	04002

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103.	Location of lead wire slot.....Gage		04003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7).....Visual		04004

4.3.2.5 Ground strap and R. F. Filter Assembly (see dwg. 9248655 covering a detail of dwg. 9248649).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Component missing or damaged.....	Visual	05001
102.	Wire lead bent or distorted.....	Visual	05002
103.	Cement missing or incomplete.....	Visual	05003
Minor:	AQL 1.00 percent		
201.	Wire length, short end.....	Gage	05004
202.	Evidence of poor workmanship (see 3.7).....	Visual	05005

4.3.2.6 R. F. Filter and Plug Assembly (see dwg. 9248649).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Distance between bridge leads.....	Gage	06001
102.	Component missing or damaged.....	Visual	06002
103.	Ground strap dented or damaged....	Visual	06003
Minor:	AQL 1.50 percent		
201.	Length of lead wires at bridge end, (min).....	Gage	06004
202.	Length of lead wires opposite bridge end, min.....	Gage	06005
203.	Evidence of poor workmanship (see 3.7).....	Visual	06006

4.3.2.7 Bridge and Plug Assembly (see dwg. 9248644 covering a detail of dwg. 9248640).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Wire and bridge visible through spot charge	Visual	07001
102.	Crack or void in spot charge	Visual	07002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	07003

4.3.2.8 Ferrule (see dwg. 9248641 covering a detail of dwg. 9248640).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Outside diameter, min	Gage	08001
Minor:	AQL 1.50 percent		
201.	Length, min	Gage	08002
202.	Inside diameter, min	Gage	08003
203.	Evidence of poor workmanship (see 3.7)	Visual	08004

4.3.2.9 Assembly, prior to inserting priming charge (see dwg. 9248640).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Weight of base charge	Balance	09001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	09002

4.3.2.10 Assembly, prior to inserting bridge and plug assembly (see dwg. 9248640).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Priming charge missing	Visual	10001
102.	Ferrule missing	Visual	10002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	10003

4.3.2.11 Assembly (see dwg. 9248640).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Lead wires not shunted	Visual	11001
2.	Cup cracked or split	Visual	11002
Major:	AQL 0.65 percent		
101.	Length	Gage	11003
102.	Diameter	Gage	11004
103.	Crimp not full 360 degrees or not flat against plug	Visual	11005
104.	Joint seal missing or incomplete	Visual	11006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	11007

4.3.2.12 Carton, prior to sealing (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Incorrect number of detonators	Visual	12001
102.	Detonators in bundle not completely encased in foil	Visual	12002
103.	Packet assembly torn or insecurely closed (when applicable)	Visual	12003

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Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	12004
202.	Package marking missing, incorrect or illegible	Visual	12005

4.3.2.13 Carton, sealed (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Carton damaged to extent that contents are exposed or liable to become exposed	Visual	13001
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect or illegible	Visual	13002

4.3.2.14 Sealed carton in sealed barrier bag (see Section 5).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Bag punctured or separation of heat seal	Visual	14001
Minor:	AQL 0.65 percent		
201.	Marking missing, incorrect, or illegible	Visual	14002

4.3.3 Testing.-Unless otherwise specified herein, the sampling plans for the following tests shall be in accordance with Standard MIL-STD-105.

4.3.3.1 Moisture content (see 3.3) - Major defects.

TABLE I

Materials	Code No.
Lead azide	11009
PETN	11010

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The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification at least one sample of each material from each eight hours production of detonators. Composite samples shall not be used. If the moisture content of a sample exceeds the requirements, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.3.2 Pull test of lead wires (see dwg. 9248649) Major Defect - Code No. 06007.-The sampling plan for this test shall be in accordance with code letter G and an AQL of 0.40 percent. Any plug assembly that fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4. All plug assemblies subjected to this test shall be scrapped.

4.3.3.3 Resistance test between lead wires (see dwg. 9248649) - Major defect - Code No. 06008.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.3 using equipment in accordance with 4.3.4. Any assembly that fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.3.3.4 Dielectric strength test (see dwg. 9248649) - Major defect - Code No. 06009.-The sampling plan for this test shall be in accordance with code letter G and an AQL of 0.40 percent. Any plug assembly that fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4. All plug assemblies subjected to this test shall be scrapped.

4.3.3.5 R. F. Attenuator test (see dwg. 9248649) - Critical defect - Code No. 06010.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.5 using equipment in accordance with 4.3.4. Any assembly that fails to comply with the applicable requirement shall be classed defective and removed from the lot.

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4.3.3.6 Pulse delay - Major defect (see dwg. 9248649) - Code No. 06011.-The sampling plan for this test shall be in accordance with code letter K and an AQL of 0.40 percent. Any assembly that fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4. Samples complying with the requirement shall be returned to the lot.

4.3.3.7 Resistance of lead circuit (see dwg. 9248640) - Major defect - Code No. 11011.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.8. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.3.8 Breakdown resistance test between the shorted lead wires and the cup (see dwg. 9248640) - Major defect - Code No. 11012.-This test shall be performed 100 percent in accordance with the procedure specified in 4.4.9. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.3.3.9 Functioning and waterproofness (see 3.4 and 3.5).

4.3.3.9.1 Sampling.-One hundred and eighty-five (185) detonator assemblies shall be selected from each lot for test. The sequence of testing and the number of samples to be subjected to each test is specified in Table II.

TABLE II

Test Sequence	Classification	Test Paragraph	Number of Samples	Code No.
1. Time (see 3.4.1)	Major	4.4.7.1	35	11013
2. Waterproofness and Output (see 3.5 and 3.4.2)	Major	4.4.7.2	50	11014
3. Duds (see 3.4.3)	Major	4.4.7.3	*100 Less time & output duds	11015

*Duds that occur in the time or output phase of testing shall be recorded and replaced from this sample.

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4.3.3.9.2 Rejection.-The lot shall be rejected if any of the following test results occur:

a. Time.-The mean functioning time ($\overline{X}$) plus the product of 2.27 and the standard deviation (s) (see 6.5) exceeds one hundred twenty (120) microseconds.

b. Waterproofness and Output.-One or more detonator assemblies fail to produce a perforation of the required size in the lead disc.

c. Duds.-The total number of detonator assemblies that fail to fire on the first attempt (dud) in the time, waterproofness and output, and dud tests equals or exceeds two (2).

4.3.3.10 Environmental (see 3.6)

4.3.3.10.1 Sampling.-Eighty-seven (87) detonator assemblies shall be selected from each lot for test. The sequence of test and the number of samples to be subjected to each test shall be as specified in Table III.

TABLE III

Test	Classification	Test Paragraph	No. of Samples	Code No.
1. Lead circuit resistance, thermal shock, and vibration (see 3.6.2)	Major	4.4.10.2	*37 resist. Time	11016 11017
2. Cold Temperature (see 3.6.3)	Major	4.4.10.3	25	11018
3. Hot Temperature (see 3.6.4)	Major	4.4.10.4	25	11019

*Time calculations shall be based on 35 firings, duds shall be recorded and replacements fired.

4.3.3.10.2 Rejection.-The lot shall be rejected if any of the following functioning results occur:

a. Time after thermal shock and vibration.-The mean functioning time ($\overline{X}$) plus the product of 2.27 and the standard deviation(s) (see 6.5) exceeds one hundred twenty (120) microseconds.

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b. Dud.-The total number of detonator assemblies that fail to fire on the first attempt after environmental conditioning as specified in 1, 2, or 3 of Table III equals or exceeds one.

c. Lead circuit resistance.-The total number of detonator assemblies that fail to comply with the lead circuit resistance requirement during thermal shock and vibration conditioning equals or exceeds one.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET-9248640 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, (production and receiving inspection, overhaul, shipping storage,) identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 Lead azide.-A sample of approximately two (2) grams (gm.) of lead azide accurately weighed to the nearest .0001 gm. shall be placed in an uncovered tared Pyrex dish approximately 55 millimeter in diameter and 30 millimeters in height. The weighed sample and dish shall be dried for 1 1/2 hours in an oven maintained at 95 plus or minus 2 degrees Centigrade (C.). At the end of the drying period, the sample shall be removed from the oven and immediately placed in a desiccator containing anhydrous calcium chloride to cool for approximately 20 minutes. After cooling, calculate the percentage of moisture in the sample from the loss in weight.

4.4.1.2 PETN.-The moisture content of the PETN shall be determined as specified in 4.4.1.1.

4.4.2 Pull test of lead wires.-This test shall be performed using the equipment specified in 4.3.4. After application of the required force to each lead wire, observation shall be made for slippage at the entry point of the wires into the plug and on the opposite face of the plug. Any unit exhibiting any slippage whatsoever, shall be classed defective.

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4.4.3 Resistance between lead wires.-The plug assembly shall be connected to the applicable test fixture specified in 4.3.4 and the voltage as specified on the applicable drawing shall be applied for the specified time. If the resistance between the lead wires is less than that specified on the applicable drawing the plug assembly shall be classed defective.

4.4.4 Dielectric strengh test.-The plug assembly shall be connected to the applicable test fixture specified in 4.3.4 and the voltage as specified on the applicable drawing shall be applied for the specified time. Any plug assembly failing to meet the specified requirement shall be classed defective.

4.4.5 R. F. Attenuator.-The plug assembly shall be connected to the applicable test fixture specified in 4.3.4 and the attenuator measured as specified. Any plug assembly failing to meet the specified requirement shall be classed defective.

4.4.6 Pulse delay.-The plug assembly shall be connected to the applicable test fixture specified in 4.3.4 and the pulse delay measured as specified. Any plug assembly failing to meet the specified requirements shall be classed defective.

4.4.7 Functioning.-The detonator assemblies shall be temperature conditioned at 70 plus or minus 10 degrees Fahrenheit (F.) for two (2) hours min. prior to performing the functioning tests.

4.4.7.1 Time.-The detonator assemblies shall be assembled to the test fixture specified in 4.3.4 and fired by applying the energy from a 16 microfarad capacitor, max. charged to 2.5 volts max. The elapsed time between application of the energy and detonator functioning shall be measured. Detonators that fail to fire shall be recorded and the replacement tested.

4.4.7.2 Waterproofness and Output.-The detonator assemblies shall be immersed in water at a two (2) to three (3) inch depth for forty-eight (48) hours min. The water temperature shall be maintained at seventy (70) plus or minus 10 degrees F. for the immersion period. At the end of the immersion time, the detonator, assemblies shall be removed from the water, assembled into the test fixture and fired by applying the same energy specified in 4.4.7.1. The lead disc perforation diameter shall be measured to determine compliance with the requirement.

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4.4.7.3 Duds.-The detonator assemblies shall be assembled in the test fixture specified in 4.3.4 and fired as specified in 4.4.7.1. Each detonator assembly shall be observed to determine compliance with the requirement.

4.4.8 Resistance of lead circuit.-Prior to twisting the lead wires together, the resistance of the lead circuit shall be measured with equipment specified in 4.3.4 to determine compliance with the requirement. The output current in the inspection equipment shall not exceed one (1) milliampere.

4.4.9 Break Down resistance between shorted lead wires and cup.-A min. potential of 500 volts, 60 cycle root mean square (r.m.s.) shall be applied for not less than one nor more than two seconds between the shorted lead wires and the cup. Equipment specified in 4.3.4 shall be used to determine compliance with the requirement.

4.4.10 Environmental tests

4.4.10.1 Temperature and humidity.-The detonator assemblies shall be tested in accordance with test 105 of MIL-STD-331. After the test thirty-five (35) of the assemblies shall be functioned as specified in 4.4.7.1 and the times recorded.

4.4.10.2 Lead circuit resistance, thermal shock and vibration.-The detonator assemblies shall be tested as follows in the sequence specified, using equipment in accordance with 4.3.4:

a. Measure resistance of lead circuit in accordance with 4.4.8.

b. Condition for 2 hours min. at minus 65 plus or minus 4 degress F.

c. Remove from cold chamber and measure lead circuit resistance in accordance with 4.4.8.

d. Condition for 2 hours min. at plus 165 plus or minus 4 degrees F.

e. Remove from hot chamber and measure lead circuit resistance in accordance with 4.4.8.

f. Vibrate for fifteen (15) minutes in a sweep covering frequency range of 25-500-25 cycles per second. Vibration amplitude shall be in accordance with Table II of MIL-STD-331 test method 104.

g. Measure resistance of lead circuit in accordance with 4.4.8.

h. The detonator assemblies shall be functioned as specified in 4.4.7.1 and the times recorded.

4.4.10.3 Cold temperature.-The detonator assemblies shall be conditioned at plus -65 plus or minus 4 degrees F. for two (2) hours min. using equipment specified in 4.3.4. The detonator assemblies maintained at the conditioning temperature shall be functioned as specified in 4.4.7.3.

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4.4.10.4 Hot temperature.-The detonator assemblies shall be conditoned at plus 165 or minus 4 degrees F. for two (2) hours min. using equipment specified in 4.3.4. The detonator assemblies maintained at the conditioning temperature shall be functioned as specified in 4.4.7.3.

4.4.10.5 Jolt.-The test shall be performed as specified in test 101 of MIL-STD-331. The detonator assembly shall be examined after the test to determine compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level B

5.1.1.1 Rosette wrap.-Twenty-five (25) detonators with twisted lead wires out straight shall be made into a bundle using the lead wires of one detonator to secure the bundle. Each bundle shall be wrapped in approximately an eight (8) by eight (8) by .0015 inch thick size sheet of aluminum foil complying with Specification MIL-A-148, so that all detonators and lead wires in a bundle are completely encased in the foil. Five (5) bundle of detonators (125 detonators) shall be packed in a folded carton set-up box, or comparable container, made of material with a minimum board thickness of 0.03 inch. When destined for shipment and uncovered storage in excess of 180 days from the loading date shown on the packing box to the date of assembly into the end items, the following shall be included: Each carton, set-up box, or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E, of Specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two subsequent seals, if required. The seal shall comply with the heat seal requirements of Specification MIL-P-116.

5.1.1.2 Pocket assembly, alternate pack.-The detonators shall be packed in pocket assemblies manufactured in accordance with Figure 1 (as applicable) made from .0015 inch thick aluminum foil complying with Specification MIL-A-148. An adhesive seal shall be used to bond the faces of the aluminum foil together to form five (5) pockets. The strength

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of the seal shall be sufficient to allow insertion of the dowel (see Figure 1) without separation of the seal. The size of each pocket between seals shall not vary by more than 1/16 inch. Each pocket shall be opened to form a cylinder using a smooth dowel with rounded end (see Figure 1). Any alternate method may be used. One M78E1 Electric Detonator assembly shall be inserted detonator end down, in each pocket. The lip of the pocket assembly shall be folded over to close the pouch. Twenty-five (25) closed pocket assemblies containing a total of 125 detonators shall be packed in a folded carton, set-up box, or comparable container made of material with a minimum board thickness of 0.03 inch. If shipment and uncovered storage is to exceed 180 days, the detonator container will be protected as specified in 5.1.1.1.

5.1.2 Level C.-Unless otherwise specified, the M78E1 Electric Detonators shall be preserved and packaged as specified in 5.1.1 except that the barrier bag is not required.

5.2 Packing

5.2.1 Level B.-A suitable number of containers each containing 125 detonator assemblies shall be packed in an outer case in accordance with Code of Federal Regulations 49 CFR for Transportation of Explosives and Other Dangerous Articles, paragraph 173.66.

5.2.2 Level C.-Unless otherwise specified M78E1 Electric Detonators shall be packed as specified in 5.2.1.

5.3 Marking.-Each carton shall be printed, stamped or labelled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. 125 Detonator, Electric, M78E1
Lot No. Date Packed (Mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a min. of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to Specification TT-I-559.

c. Printing and labelling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with dwg. 8796522. The Dot designation shall be (see Note): Electric Blasting Caps - Handle Carefully.

e. Note: "The number of caps packed in one outer pack shall be shown in the marking".

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspection listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection equipment lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicated the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Designs" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

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6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list reference mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless other specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a refernce to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission

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shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of test data and ammunition data cards.- In addition to the normal distribution of records, for loaded item procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to the following:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND5
SMUPA-DE
Dover, New Jersey 07801

6.5 Calculation of standard deviation.-The following formula or equivalent shall be used in calculating the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(x-\bar{X})^2}{n-1}}$$

where:

s = sample standard deviation
X = numerical value of individual observations
$\bar{X}$ = average value of the sample observations
n = sample size

6.6 Check test for possible deterioration.-The following test is designed to prevent major ammunition items being loaded with detonators that may have deteriorated. This test shall be performed on detonators from each lot which has been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require a check test, the contracting officer shall insert into the contract a check test performance provision. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the deterioration check test. Two hundred (200) detonator assemblies shall be selected and tests as follows:

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6.6.1 Function time and duds.

6.6.1.1 Sampling.-Forty (40) detonators shall be subjected to the test specified in 4.4.7.1. Observation shall be made for the functioning time, and all duds that may occur will be recorded. Duds will be replaced by detonators from the sample of 200.

6.6.1.2 Rejection.-The lot shall be rejected if the mean functioning time ($\bar{X}$) plus 2.03 times the standard deviation(s) exceeds 120 microseconds. Duds shall not be used in the calculation for standard deviation.

6.6.2 Waterproofness, output and duds.

6.6.2.1 Sampling.-Fifty (50) detonator assemblies shall be subjected to the test specified in 4.4.7.2. Observation shall be made of the waterproofness and output, and duds that may occur will be recorded. Duds will be replaced by detonators from the sample of 200.

6.6.2.2 Rejection.-The lot shall be rejected if two or more detonator assemblies fail to comply with the disc perforation requirement.

6.6.3 Duds

6.6.3.1 Sampling.-The remainder of the 200 detonator assemblies selected in 6.6 shall be subjected to the test specified in 4.4.7.3.

6.6.3.2 Rejection.-The lot shall be rejected if the number of detonator assemblies that failed to fire on the first attempt (duds) accumulated from 6.6.1.1, 6.6.2.1, and 6.6.3.1 equals or exceeds four (4).

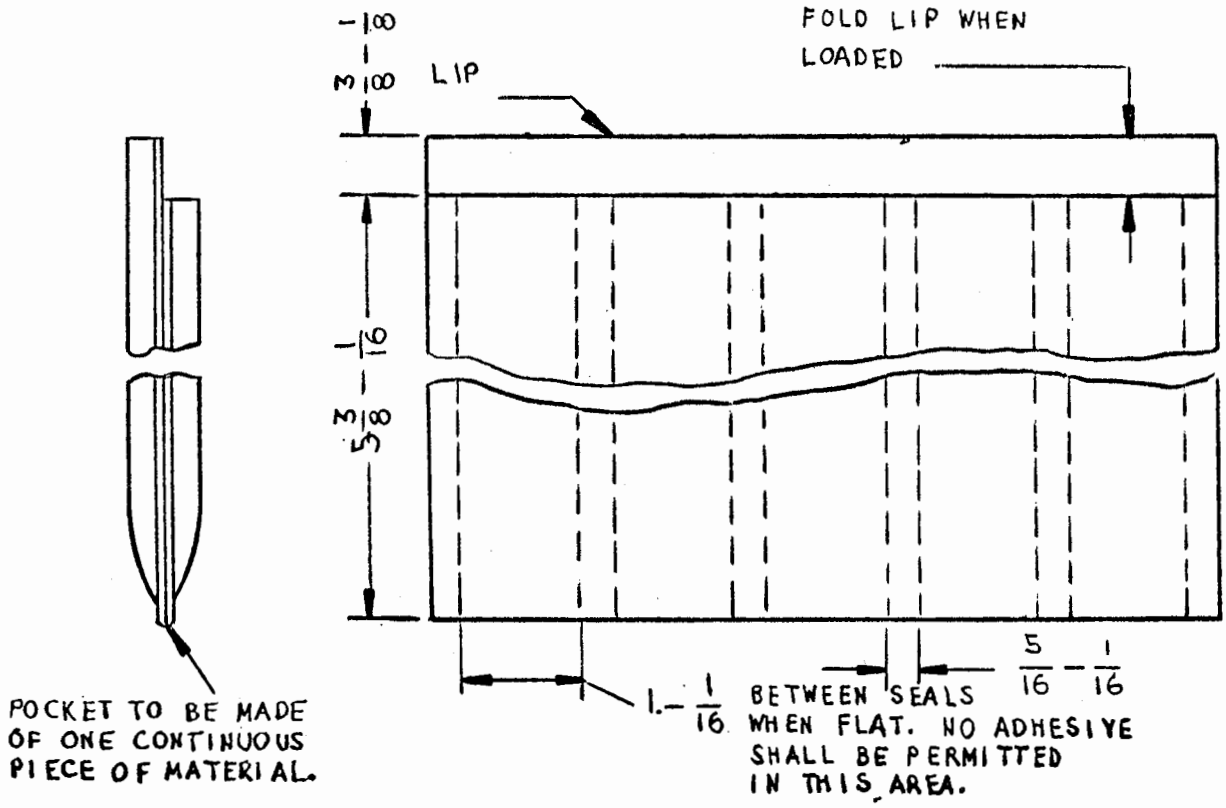
6.7 Supersession data.-This specification included the requirements of PA-PD-2965, dated 26 May 1967.

Custodian:
Army-MU

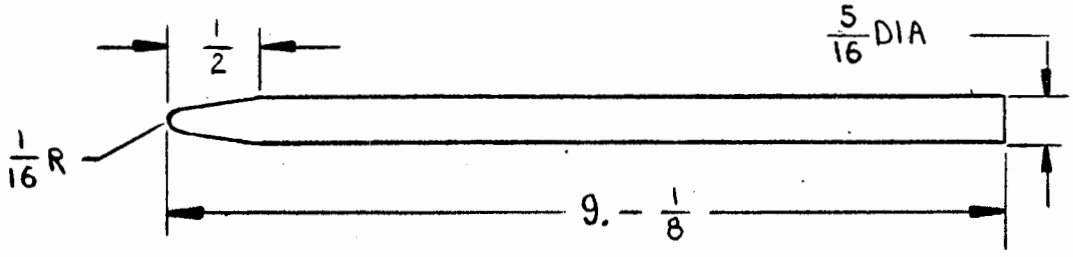
Preparing activity:
Army-MU

Project No. 1375-A-773.

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POCKET ASSEMBLY



DOWELL SHALL BE SMOOTH WITH NO SHARP CORNERS.

DOWEL

FIGURE 1

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